CHILANGA CEMENT

A MEMBER OF HUAXIN

SCOPE OF WORK: REPLACEMENT OF CEMENT MILL 1ST & 2ND CHAMBER DIAPHRAGM LINERS

JANUARY

2026

Chilanga cement Zambia Plc
Chilanga Plant

Scope of Works

1.0 INTRODUCTION

1. Introduction

Chilanga Cement Zambia, CHL2 Plant, operates a cement ball mill located on the northern side of the administration block. The ball mill, measuring 4,660 mm in diameter and 15,350 mm in length, is a critical component of the cement production process. It features an intermediate frame that divides the mill into two distinct grinding chambers—the first chamber for primary coarse grinding and the second chamber for fine grinding.

The purpose of this tender is to solicit proposals from qualified contractors for the **removal and replacement of the first and second chamber diaphragm liners**. The work includes the careful dismantling of the worn-out liners, installation of new liners according to manufacturer specifications, and ensuring proper alignment and fitment to maintain optimal grinding efficiency.

2. Safety

Safety is key consideration for Lafarge and the contractor who will be picked to carry out works under the Scope details on the next page will need to:

a) Abide by Lafarge safety requirements especially as regards to PPE and general SOPs. Minimum requirement for PPE is Safety boot with toe cap, work suit with reflective material (Orange work suits with 75 % reflectors), hard hat, overlapping goggles with seals and dust respirator. See attached type of dust seal goggles and 75% reflective work suit.

b) All works above 1.8m from floor or any platform have to be executed using Lafarge Work at Height (WAH) procedures which are outlined during contractor

induction. Contractor is required to have full body Harnesses for all persons that are going to work at Height.

c) Documented Risk Assessment has to be done at the beginning of every task on site. As work progresses, the team will be required to continuous reassess the risks associated to the task factoring the task changing conditions.

d) Contractor must get himself aware of Lafarge safety standards (CSM) before submitting offer.

e) The area needs to be barricaded before starting a job.

3. Scope of Work

3.1 Pre-Shutdown Activities

- Review approved drawings, liner layout, and OEM instructions.
- Conduct site inspection and familiarization.
- Prepare method statement, risk assessment (JSA), and lifting plan.
- Conduct toolbox talks prior to commencement of work.

3.2 Isolation and Mill Entry

1. Carry out full lock-out try-out and tag-out (LOTOTO) of the cement mill.Personal locks and tags must be strictly observed in accordance with the one key–one lock principle. Only approved colours—green, blue, or yellow—are permitted.

- Open manholes and provide safe access for confined space entry.
- Ensure adequate lighting, ventilation, and dust suppression.

3.3 Dismantling Works

- Dismantle existing diaphragm liners, screens, and fastening bolts on:
- First chamber side
- Second chamber side
- Lower and remove dismantled components safely from the mill.

3.4 Diaphragm Liners Installation (Chilanga Cement-Supplied)

First Chamber Side

Install 115 diaphragm liners arranged in five (5) rows:

- Row 1: 23 liners
- Row 2: 23 liners
- Row 3: 23 liners
- Row 4: 23 liners
- Row 5: 23 liners

Second Chamber Side

- Install 92 diaphragm liners arranged in four (4) rows:
- Row 1: 23 liners
- Row 2: 23 liners
- Row 3: 23 liners
- Row 4: 23 liners

Ensure correct discharge opening and ventilation arrangement.

3.5 Bolting and Fastening

- Install chilanga cement -supplied bolts, washers, and locking devices.
- Tighten all bolts to specified torque values 1500nM .
- Ensure all fasteners are securely locked and compliant with OEM standards.

3.6 Inspection and Quality Control

- Inspect diaphragm assembly for correct fitment and alignment.
- Verify that airflow slots and material passages are unobstructed.
- Rectify any misalignment identified during installation.

3.7 Mill Closure and Handover

- Remove all tools, materials, and debris from inside the mill.
- Close and secure all manholes.
- Assist with mill rotation checks if required.
- Submit completion report and hand over to operations.

4. Materials

Chilanga cement to supply:

- Diaphragm liners
- M33X2 Bolts, nuts, washers, and locking devices
- Gas
- 14 LBS Hammers
- Gas detector
- Chain blocks
- Slings
- Welding electrodes
- Cutting disc

Contractor to supply:

- Manpower, tools, lifting equipment, and consumables required for installation.

Hand Tools

- Hammer: ~5 kg, professional quality, good working condition.
- Peen hammer: ~2 kg.

- Wedges: suitable for liner and material adjustments.
- Crowbars: heavy-duty, for leverage work.

Requirement: Contractor must provide at least 2 qualified rigger per shift.

Documentation: Hard copies of rigger certificates must be submitted to the Planning Office prior to commencement.

Power Tools & Welding Equipment

- Grinding machine.
- Welding machine
- Cutting torch.
- Bolting & Torque Tools
- Pnematic wrench

5. Safety, Health & Environment

- Comply with site safety rules and confined space requirements.
- Provide competent supervision and certified artisans.
- Ensure proper handling and disposal of removed liners and scrap.
- Maintain housekeeping throughout the work.

6. Exclusions

- Supply of diaphragm liners and bolts.
- Major structural repairs to diaphragm frames or mill shell.

- Grinding media supply or optimization.
- Electrical or instrumentation works.
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7. Deliverables

- Completed replacement of diaphragm liners (1st & 2nd chamber).
- Installation and inspection report.
- Bolt torque records.
- Final handover and sign-off.

The offers have to be sent to the attention of:

Procurement Specialist

Chipadzuwa chalwe

Chilanga Cement PLC

Kafue Road | P.O.Box: 32639 | Lusaka | Zambia

Office: +260 211 376 898 | Mobile: +260779311233

Email: chipadzuwa.chalwe@huaxincem.com